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Ministry of Education & Higher Education
Palestine Technical College - Deir El-Balah

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كلية فلسطين التقنية - دير البلح

Information and Communication Technology

Computer Department

"Himmah"

A Virtual Learning Platform for Palestinian 12th grade Students in Gaza

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DEDICATION

To those who inhabited our souls before our hearts did, to all who have traveled with us along these paths of pain and hope, we offer you the fruit of this labor.

We dedicate it to our wounded homeland, Palestine, which teaches its children that even in its war, destruction, bombardment, and displacement, there is hope born from below the rubble, and knowledge is the weapon of survival and the spirit of resistance.

We dedicate it to the spirits of the martyrs who departed from us in body but continue to live among us, to inspire us to carry on the path and to raise the banner of knowledge as they raised the banner of the homeland. We dedicate it to our mothers and fathers, who during our lives with them, endured the desolation of days and darkness of nights with us, who were the loving prayer that kept us safe, the tear that hid the pain, and the smile that gave us strength.

To our siblings, relatives, and those we love who suffered with us fear and displacement, and who created warmth and safety of family even when it was crushed by rubble, we owe you our enduring gratitude. To each friend, teacher, and anyone who believed in person, and took the time to offer a word of kindness, or a real prayer or a bow of support, we dedicate this work.

This isn't an academic endeavor, this work is a testament of life: a testament that says, despite war and bombardment, despite displacement and destruction, we held onto the dream, we got the work done and withstood the storms until we came here.

This becomes evidence that will creates the monumental, that the light of knowledge can never be extinguished by violence and uncertainty of war, and that the people who love their homeland will always cultivate a new seed of tomorrow on the ruins that were just leveled.

ACKNOWLEDGMENT

All praised is to Allah who granted us strength, endurance, and will to now reach a phase in our academic lives.

Peace and blessings of Allah be upon our Prophet Muhammad, his family, and his companions. As we reach the completion of this phase of our college lives, we would like to express our sincere appreciation to everyone that helped us along the way.

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This completion is also dedicated to everyone who stood by us and believed in us.

DECLARATION

I hereby certify that this material, which I now submit for assessment on the program of study leading to the award of Bachelor of Web Application Technology and Information Security, is entirely my own work, that I have exercised reasonable care to ensure that the work is original, and does not to the best of my knowledge breach any law of copyright, and has not been taken from the work of others save and to the extent that such work has been cited and acknowledged within the text of my work.

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ABSTRACT

The ongoing conflict and humanitarian crisis in Gaza have placed an unprecedented strain on the educational infrastructure, disrupting the schooling of thousands of students. With schools damaged and many students displaced, the need for alternative modes of education has never been more urgent. This project proposes the development of an innovative e-learning platform designed specifically for 12th-grade students in Gaza, offering a comprehensive and accessible digital solution to continue their education despite these dire circumstances.

The platform will provide video lessons, interactive quizzes, downloadable resources (PDFs), and direct communication with teachers. This will ensure that students can study and prepare for the Palestinian Tawjihi exams, regardless of the challenges they face. The platform will also foster a collaborative learning environment, where students can engage with teachers and peers, ensuring continuity in their academic journey. By leveraging modern web technologies and tailored educational content, this project aims to bridge the gap caused by physical displacement and provide an effective, flexible, and scalable solution for education in Gaza.

Keywords: Educational content, Access to education, Teacher-student communication, PDF resources, Video lessons, Gaza, Humanitarian crisis, 12th-grade students, Tawjihi exams, Interactive learning, Academic success, Flexible learning environment.

Chapter 1

Introduction

INTRODUCTION

The educational system in Gaza has faced repeated setbacks in recent years due to the ongoing Israeli-Palestinian conflict. Schools have been damaged or destroyed, and many students have been displaced, further exacerbating an already fragile situation. In this challenging context, the traditional methods of classroom learning have become nearly impossible for many students. As a result, these young learners particularly those preparing for the crucial 12th-grade Tawjihi exams are at risk of falling behind, which could have long-term consequences for their academic and professional futures.

In light of these challenges, the need for a flexible, online learning platform is critical. E-learning has proven to be a vital tool in providing education in crisis zones worldwide, offering both accessibility and scalability. The platform envisioned in this project aims to replace traditional schooling with an innovative digital environment, where students can access quality educational content, engage in interactive activities, and receive support from teachers—all from the safety and comfort of their homes or shelters.

The proposed platform will be tailored specifically to the needs of Palestinian students in Gaza. It will focus on the 12th-grade curriculum across multiple tracks—scientific, literary, and entrepreneurial—ensuring that students from all fields of study have equal access to educational resources and tools to succeed. The platform will not only support students' academic needs but also foster a sense of community and normalcy during a time of great uncertainty.

1.1 Problem statement

The ongoing conflict in Gaza has led to widespread destruction of educational facilities, displacement of students, and a significant disruption to the academic calendar. Thousands of students are currently unable to attend school, either due to the physical destruction of schools or the lack of a safe environment for learning.

For 12th-grade students, this disruption is particularly detrimental, as they face the high-stakes Tawjihi exams that determine their future access to higher education and employment opportunities.

In addition to the physical destruction, the mental and emotional toll on students has been immense. Many have witnessed the loss of loved ones, homes, and entire communities. For these students, returning to a classroom environment is no longer a viable option, leaving them with limited educational opportunities. Online education, although not without its challenges, offers a viable solution. However, the current digital divide marked by limited access to technology, unstable internet connections, and lack of educational resources has made it difficult for many students to benefit from e-learning.

Moreover, the absence of structured support from teachers, who are also affected by the same disruptions, further compounds the problem. Students are left isolated without proper guidance, making it even harder for them to engage with the curriculum or seek clarification when needed.

This project seeks to address these issues by developing a dedicated, user-friendly e-learning platform that:

1-Fills the gap in formal education by offering a fully digital and comprehensive alternative to traditional schooling.

2-Provides equitable access to learning materials and resources, regardless of geographic location or the availability of physical infrastructure.

3-Offers direct communication between students and teachers, ensuring that students can receive the support they need during these difficult times.

4-Facilitates learning through engaging, interactive content, making the educational experience more adaptable to the diverse needs of students.

This platform will empower Gaza's 12th-grade students to continue their education uninterrupted, giving them the tools, they need to succeed in their exams and, by extension, secure their futures.

1.2 Objectives

In this section, we show both main and specific objectives of our work.

1.2.1 Main objective

The primary objectives of an educational platform designed for Palestinian Tawjihi students should focus on enhancing educational access, improving academic

performance, and supporting the unique needs of students preparing for the Tawjihi exam. Here are the main objectives:

1) Provide Comprehensive and Relevant Learning Resources:

Objective: Ensure the platform offers a complete range of study materials aligned with the Tawjihi curriculum.

Action: The platform will feature textbooks, study guides, interactive lessons, videos, and practice exams covering all subjects of the Tawjihi, including Arabic, English, mathematics, sciences, social studies, and Islamic education.

Outcome: Students will have access to quality, localized content that prepares them thoroughly for the exam, promoting comprehensive understanding rather than rote memorization.

2) Bridge the Educational Gap Across Palestine:

Objective: Provide equal access to quality education for all students, regardless of their geographic location or socio-economic status.

Action: Ensure the platform is accessible via smartphones, tablets, and computers, with mobile-first design for areas with limited access to traditional computer labs. Provide free or affordable access to core learning materials, with premium content available for those who can afford it.

Outcome: Students in remote or underserved areas, as well as those from low-income backgrounds, will be able to access the same high-quality educational resources, leveling the playing field for all students.

3) Enhance Teacher Engagement:

Objective: Engage teachers in the educational process to ensure a supportive learning environment for students.

Action: The platform will offer features for teachers to monitor student progress, assign personalized homework, and provide additional resources.

Outcome: Continuous collaboration between students and teachers, will create a more holistic support system, ensuring that students receive the guidance they need to succeed.

1.2.2 Specific objectives

Here are the specific objectives of our project:

1) Provide Access to Comprehensive Study Materials:

Objective: Offer a wide array of subject-specific study resources that align with the Palestinian Tawjihi curriculum.

Action:

- Upload textbooks, study guides, lesson plans, and detailed notes for each subject.
- Include video tutorials, audio lectures, and visual aids (e.g., diagrams, animations) to enhance learning.
- Update resources regularly to match curriculum changes and new exam formats.

Outcome: Students have direct access to a variety of materials in one platform, reducing the need for multiple sources and streamlining their study process.

2) Provide Practice Tests:

Objective: Offer practice exams and quizzes designed to replicate the format and difficulty of the Tawjihi exams.

Action:

- Provide a library of past Tawjihi exams and generate randomized quizzes.
- Include time limits to simulate real exam conditions and promote time management.
- Provide instant feedback after each mock exam, highlighting correct and incorrect answers with explanations.

Outcome: Students can assess their performance, identify weaknesses, and improve

3)Promote Real-Time Progress Monitoring for Students and Teachers:

Objective: Ensure that students and teachers can track learning progress and performance in real-time.

Action:

- Implement a dashboard that tracks student activities, including completed lessons, quizzes, and exams.

- Provide progress reports for teachers to monitor and support students' learning.

- Include notifications for upcoming exams, assignments, and areas requiring improvement.

Outcome: Real-time tracking helps all stakeholders stay informed and intervene when necessary, ensuring that students remain on track for success.

5) Enhance Performance and Responsiveness

Objective: Improve the platform's speed, efficiency, and responsiveness to ensure a smooth user experience under all network and device conditions.

Action:

- Implement caching and lazy loading techniques to accelerate page rendering and reduce loading times.
- Optimize images and other media files to minimize bandwidth usage.
- Ensure efficient asset loading and content delivery to maintain performance consistency across different catalogue sizes and devices.

Outcome: The platform will deliver faster load times, smoother navigation, and a more responsive interface, ensuring users enjoy a seamless experience even in areas with limited internet connectivity.

1.3 Importance of the project

Now we discuss the main points that make our project important

1)Enhanced Learning Experience:

- Interactive Learning: Offers engaging multimedia content like videos, simulations, and interactive quizzes.

- Personalized Learning: Provides tailored learning paths based on individual student needs and progress.
- Real-time Feedback: Delivers immediate feedback on assignments and quizzes to facilitate learning.

2)Improved Academic Performance:

- Comprehensive Study Materials: Provides access to digital textbooks, and other study materials.
- Practice Exams and Mock Tests: Offers ample opportunities to practice prepare for the Tawjihi exams.

3)Equitable Access to Education:

- Remote Learning: Enables students to access educational resources regardless of their location.
- Accessibility: Ensures inclusivity by designing the platform to be accessible to students with disabilities.
- Affordability: Offers affordable or free access to a wider range of students.

4)Data-Driven Insights:

- Track Student Progress: Monitors student performance and identifies areas for improvement.
- Inform Pedagogical Practices: Uses data to optimize teaching strategies.
- Evaluate Platform Effectiveness: Analyzes usage data to refine the platform.

1.4 Scope and Limitations of the Project

In Scope

The scope of Himmah-SYS includes the development of a **web-based and mobile-accessible e-learning platform** for Tawjihi students and teachers. The system shall provide features to facilitate learning, teaching, communication, and progress tracking. The system will be developed using web technologies such as **HTML, CSS, JavaScript, and PHP**.

Key features include:

User registration and login: Allow students and teachers to create accounts and securely log in. Role-based access ensures each user sees functionalities relevant to their role.

Content management: Teachers can upload, update, and delete lessons, quizzes, and study materials (PDFs, videos, exercises). Students can access and download these resources.

Interactive quizzes and assessments: Students can take quizzes with instant feedback, and their progress will be recorded in the system.

Student progress tracking: Dashboards show students' completed lessons, quiz scores, and overall performance. Teachers can also view reports.

Communication tools: Messaging and email forums for student-teacher interaction and collaborative learning.

Notifications: Automated alerts for upcoming lessons, assignments, or important announcements.

Admin management: Administrators can manage users, subjects, and content, as well as monitor system activity to ensure smooth operation.

Out of Scope

The following features are **not included in Himmah-SYS** for the current phase:

Official exam administration: Himmah-SYS will not conduct or replace formal Tawjihi exams or issue official certifications.

Advanced AI-based learning personalization: The platform will not include adaptive AI algorithms for personalized learning paths in this version.

Offline content delivery: Full offline functionality for lessons and quizzes will not be supported initially; internet access is required.

Third-party integration for gamification or analytics: Integration with external gamification tools or advanced analytics platforms is not included.

Mobile applications: Dedicated iOS or Android mobile apps are not part of the current scope; users will access the platform via web browsers.

Chapter 2

Related work

Related works

2.1 State of the art/review of related works

Gaza's education system has been severely impacted by ongoing conflicts, which have disrupted traditional learning methods and forced schools to close for extended specifically periods. In this context, developing an innovative e-learning platform designed for 12th-grade students becomes essential to ensure continuity in education. This section reviews previous studies and research on the development of .e-learning platforms

Presentation of Previous Studies with Sources:

1)"The Role of E-Learning in Enhancing Education in Gaza during the COVID-19 Pandemic" (Hassan & Shteiwi, 2020):

A study examined the role of e-learning in improving education in Gaza during the COVID-19 pandemic. The study found that the pandemic has accelerated the shift to e-learning in many schools in Gaza. However, the study highlighted the significant challenges faced by students and teachers in using e-learning platforms, including poor internet connectivity and insufficient devices. On the other hand, the study showed that e-learning provided opportunities to expand access to education, especially in light of school closures.

2)"Challenges and Opportunities of E-Learning in Gaza" (Al-Sabbagh and Al-Saadi, 2019):

This study discussed the challenges and opportunities facing e-learning in Gaza. The study confirmed that the biggest challenges are the lack of modern devices such as laptops and tablets, in addition to the continuous power outages and difficulty connecting to the Internet. The study also presented the opportunities provided by e-learning such as the possibility of distance learning and access to a variety of educational resources. It also recommended the need to improve the Internet infrastructure and provide appropriate devices for students.

3)[Johnson & Brown, 2019] study:

showed that e-learning allows for greater interaction between students and content, as students can choose the appropriate time and place to learn the subject matter. The presence of interactive tests also contributed to enhancing students' understanding and motivation.

4)[Harrison & Taylor, 2020] study:

focused on the importance of personalizing content based on student data and performance, as artificial intelligence techniques were used to personalize lessons and exercises based on each student's level. The study showed that personalizing content can increase the effectiveness of learning.

5)[Liu & Wang, 2021] study:

highlighted the role of multimedia such as educational videos and animated illustrations in making study materials more attractive to students. It showed that multimedia helps in communicating scientific concepts faster and more understandable.

6)[Lee et al., 2021] study:

discussed the importance of designing easy and convenient user interfaces for high school students. It showed that easy and direct interaction with content is the most influential factor on students' academic achievement.

7)"توجيهي بال" (Tawjihipal) Platform

This is a Palestinian online educational platform specifically designed for Tawjihi students, offering interactive and localized resources. The platform features subject-matter lessons, video tutorials, continuous online classes solving previous years' questions, student accounts, and forums for communication and peer support.

Key strengths include its focus on the Palestinian Tawjihi audience, mobile accessibility, and integration of psychological and educational support services. However, given the context, potential limitations may include dependency on internet connectivity and device availability (which mirror the general challenges seen in Gaza and the West Bank).

8) "منصة A+ التعليمية" (Aplus.ps)

This Palestinian e-learning platform targets high school Tawjihi students and offers simplified, comprehensive explanations of curricula via video lessons, interactive files, and assessment tests.

Feature / Aspect	Tawjihipal Platform	A+ Platform (Aplus.ps)	Himmah-Tawjihi
Target Audience	Palestinian Tawjihi students	Palestinian Tawjihi students	Palestinian Tawjihi students (Grade 12)
Curriculum Alignment	Aligned with Palestinian Tawjihi curriculum	Aligned with Palestinian Tawjihi curriculum	Fully aligned with Tawjihi curriculum; updated regularly according to Ministry of Education standards
Learning Content	Recorded lessons, previous exam solutions, live classes	Video lessons, exercises, and assessments	Comprehensive content: textbooks, study guides, video tutorials, interactive lessons, and practice exams
Interactivity	Live sessions and question-answer discussions	Interactive exercises and teacher communication	Adaptive learning with quizzes with instant feedback, gamified progress tracking
User Roles	Students only	Students and teachers	Students and teachers
Personalization	Limited (same content for all users)	Limited personalization through choice of lessons	Limited personalization through choice of lessons
Offline Access	Requires stable internet connection	Online only	Online only
Monitoring and Reports	Not available	Basic student tracking	Real-time dashboards for students, and teachers with detailed progress reports

Feature / Aspect	Tawjihipal Platform	A+ Platform (Aplus.ps)	Himmah-Tawjihi
Device Accessibility	Mobile and desktop	Mobile-friendly design	Cross-platform (web, mobile, tablet) with mobile-first design for accessibility
Language	Arabic	Arabic	Arabic
Community Support	Student community and discussion forums	Collaborative features for teachers and students	Teacher support and student Q&A
Additional Features	Focuses on solving past exams	Continuous tests and instructor guidance	Messenger (chat between student and teacher)
Pricing Model	Mostly paid content	Subscription-based	Free core materials + optional premium content
Performance Optimization	Standard web delivery	Optimized videos	Optimized videos
Overall Evaluation	Strong local relevance, lacks personalization	High-quality lessons, limited AI features	Comprehensive, inclusive, adaptive, and performance-optimized platform designed to fill current gaps

Table 1: Comparative Analysis of Himmah-Tawjihi and Other E-learning Systems

Chapter 3

Methodology and Requirements Analysis

Methodology and Requirements Analysis

3.1 Theories and Methods:

This theory is based on the principle that learning occurs as a result of repetition and reinforcement. In the context of technological education e-learning platforms can be used to enhance this type of learning This theory focuses on the fact that students build their knowledge through experiences and interaction with the environment educational technology such as simulation and virtual reality can support this approach. This theory reinforces the idea that learning occurs through networks and communications. The use of the Internet and social networks can greatly support this theory.

Rationale and Philosophical Assumptions:

This site aims to explore how technology can improve the quality of education and make it more effective and comprehensive We assume that technology is not just a tool, but a medium that can redefine traditional teaching methods and enhance the educational experience

Methods in Software Development Projects:

- **Information Collection:** Data will be collected using online questionnaires and interviews with teachers and students.
- **Data Analysis:** Statistical methods will be used to analyze the collected Data such as descriptive analysis and inferential analysis.
- **System Design:** The system design will include different learning models such as blended learning, collaborative learning, and relying on smooth interactive interfaces

Existing Methods:

- **Algorithms**: Applying machine learning algorithms to analyze student data and provide personalized recommendations
- **Approaches:** Adopting different educational approaches such as self-learning

Example of implementing the methodology in education:

- **Data Collection:** An electronic questionnaire was designed to assess the opinion of students and teachers about the effectiveness of using technology in education The questionnaire included questions about interaction, understanding and overall satisfaction with the educational experience.

- **Data Analysis:** The responses were analyzed using statistical programs such as SPSS The results showed that 85% of students believe that using technology helped them understand the material better, while 90% of teachers believe that technology increased student interaction in the classroom.

- **System Design:** The proposed educational system was based on the Moodle platform, or Classroom, and interactive modules such as educational videos and self-tests were added. Data analysis was also used to provide personalized recommendations for each student based on their performance

3.2 Software Tools and Equipment Requirements:

1)Integrated Development Environment (IDE):

- Visual Studio Code: A popular and easy-to-use editor that supports many programming languages
- XAMPP to run the serve

2)Database Management System (DBMS):

- MySQL: To manage student and course data

4)Testing Devices:

- Multiple Browsers: Like Chrome, Firefox, Safari to test browser compatibility
- Mobile Devices: For testing mobile device compatibility

3.3 Requirements Elicitation

Requirements elicitation is the process of identifying what the system must do and the constraints under which it must operate. For the Tawjihi educational platform, the project team employed multiple techniques to gather both functional and non-functional requirements from the target users and contextual analysis.

1) Student Survey (Google Form)

- **Purpose:** To gather quantitative and qualitative insights from Tawjihi students regarding their learning preferences, study habits, and accessibility challenges.
- **Method:** An online Google Form survey distributed to a sample of students, containing multiple-choice, rating-scale, and open-ended questions.
- **Outcome:**

  Preferred Learning Resources: 60% preferred video tutorials, 55% valued practice exercises, and 40% relied on textbooks.

  Accessibility: 45% reported unstable internet access, 30% lacked consistent device availability.

  Exam Preparation: 70% requested mock exams and performance tracking.

  Engagement: 50% favored gamification features for motivation.

  Implication: The platform should prioritize **multimedia content, adaptive learning, offline access, mobile compatibility, and gamified progress tracking**.

2) Stakeholder Interviews

- **Purpose:** To understand expectations and pain points from users involved in the educational process.
- **Participants:**

  -Tawjihi students (focus group discussions).

  -Teachers and educators responsible for curriculum delivery.

  -Parents monitoring their children's academic progress.
- **Outcome:** Key insights were collected on desired learning features, assessment needs, and interactive content formats.

3) Contextual Journey Review

- **Purpose:** To analyze typical learning workflows and identify information needs or friction points in the study process.

- **Method:** Observation and mapping of a student's learning journey, from selecting a subject to completing lessons, quizzes, and exams.
- **Outcome:** Determined the essential platform features needed to streamline the learning experience, such as **lesson sequencing, progress tracking, and interactive assessments**.

4) Prototype Walkthroughs

- **Purpose:** To validate the platform's interface design, navigation, and usability before full development.
- **Method:** Low-fidelity wireframes created for the main workflow: Home → Subject Selection → Lesson → Practice Test → Progress Dashboard.
- **Outcome:** Verified usability, clarity of labels, step count, and intuitive navigation for students, teachers, and parents.

5) Policy and Standards Review

- **Purpose:** To ensure the platform complies with legal, ethical, and educational guidelines.
- **Method:** Review of data privacy laws, user account policies, copyright compliance, and ethical standards for educational content.
- **Outcome:** Requirements were established to maintain **security, clarity, and trust** in all platform interactions.

3.4 Actors

Student (Guest/Registered)

- **Guest:** Can browse available courses and lesson summariesand view general platform information.
- **Registered Student:** Performs all guest actions plus: enrolls in courses, views full lessons and resources (PDFs/videos), takes quizzes and mock exams, tracks progress through dashboard, and messages teachers.

Teacher

- Creates and manages course content (videos, PDFs, quizzes).
- Monitors student progress and grades.

- Sends announcements, responds to student messages, and reviews quiz results.
- Can schedule live sessions and moderate discussion forums.

Administrator

- Manages user accounts (students, teachers, parents).
- Oversees course approvals, lesson publishing, and platform moderation.
- Handles system settings, reports, and maintenance.
- Monitors platform statistics and ensures policy compliance.

System (Automated Components)

- Handles user authentication and authorization.
- Sends notifications (email/in-app).
- Tracks analytics, generates performance reports, and performs scheduled backups.

External Services

- **Cloud Storage Service:** For hosting videos and downloadable PDFs.
- **Email/Notification Service:** For sending verification emails, updates, and alerts.
- **Payment Gateway :** For processing premium content or certification fees.

3.5 Functional Requirement

Admin Side

Homepage & Slider Management

- The admin must be able to upload and manage banner or promotional images (e.g., upcoming courses, announcements).
- The admin must be able to arrange the order of slider images based on priority or campaign.
- The admin must be able to edit or delete any slider image.

Category / Subject Management

- The admin must be able to create and manage main academic categories (e.g., Science, Literary, Entrepreneurial tracks).
- The admin must be able to add, edit, or delete subcategories (e.g., Physics, Arabic Literature, Business Management).
- Each course must be linked to one main category and, optionally, a subcategory.
- The admin must be able to manage course visibility (active/inactive, featured, archived).

Teacher & Student Account Management

- The admin must be able to view all registered teachers and students.
- The admin must be able to edit user roles or deactivate accounts when necessary.
- The admin must be able to view teacher profiles, assigned courses, and student enrollments.

Communication & Announcement Management

- The admin must be able to post global announcements visible to all users.
- The admin must be able to moderate discussion forums or comment sections.
- The admin must be able to review contact messages or reports submitted by users.

Exam & Assessment Oversight

- The admin must be able to view all quizzes and mock exams created by teachers.
- The admin must be able to track student participation and performance statistics.

Reporting & Analytics

- The admin must be able to access reports on platform usage (e.g., number of users, courses, quiz attempts).
- The admin must be able to view performance summaries for students and teachers.

- The admin must be able to export data as CSV or PDF for further analysis.

System Settings

- The admin must be able to manage site settings such as logo, contact info, and support email.
- The admin must be able to configure notification settings (email templates, SMS alerts).
- The admin must be able to manage backup and restore operations for the database.

Contact Us Messages

- The admin must be able to view and manage all messages sent by students or teachers through the "Contact Us" form.
- The system should display the sender's name, email, message subject, and timestamp.

System Settings / Change Password

- The admin must be able to modify general site settings such as **platform name, logo, contact email, and notification preferences**.
- The admin must be able to update **integration settings** for external services (e.g., email server, cloud storage, optional payment gateway for premium courses or certificates).
- The admin must be able to **change their account password** securely through the control panel.
- All configuration changes must be logged for audit and security purposes.

<u>Second: User Side (Student Interface)</u>

Menu Navigation

- The main menu must include links to all important site sections such as: **Home**, **Subjects/Courses**, **My Dashboard**, **Quizzes**, **Messages**, **About Us**, and **Contact Us**.
- The menu must adjust dynamically based on the user's login status (e.g., show "Login / Register" for guests, "My Account / Logout" for logged-in students).

• Subject / Category Browsing

- Students must be able to browse main academic tracks (Scientific, Literary, Entrepreneurial).
- Each main category must include relevant subcategories (e.g., Physics, Chemistry, History, Economics).
- Users can filter or search courses by subject, teacher, difficulty, or type (video lesson, PDF, quiz).

Course Listing

- A list of available courses must appear under each subject category.
- Each course card should show the **course title, teacher name, short description, and thumbnail image**.

Course Detail Page

- Students must be able to view detailed course information, including:
  - Full course description
  - Lesson list (with status: completed / in progress)
  - Teacher profile and contact link
  - Embedded video lessons and downloadable PDFs
- Each course page must also include quizzes.

Add to My Courses (Enrollment)

- Students must be able to enroll in a course (free or premium).
- Once enrolled, the course appears in the student's **Dashboard → My Courses** list.
- Progress tracking must update automatically as lessons or quizzes are completed.

Login

- Users must be able to log in using their **email and password** (or optionally via Google login if implemented).
- Passwords must be securely encrypted; login sessions should expire automatically after inactivity.

Register

- New users must be able to create an account by providing: **name and email, password.**
- A verification email must be sent to activate the account before access to full features.

Forgot Password

- Students must be able to recover their password using the email address registered with their account.
- A password reset link must be sent to the user's email, allowing them to securely create a new password.
- The link must expire after a set time for security purposes.

My Courses Page

- Students must be able to view all courses they are enrolled in.
- Each course card must display progress (e.g., "5/10 lessons completed"), next recommended lesson, and course status (active or completed).
- Students must be able to **unenroll** from a course if allowed by the admin.

Course Purchase or Enrollment (For Paid Courses)

- Students must be able to enroll in free courses instantly.
- For premium courses, students must be able to complete enrollment using integrated payment options (e.g., **PayPal**).
- Upon successful payment, the student's account should automatically gain access to the paid course.
- Payment details must be logged securely for reference.

Progress and Assessment Tracking

- Students must be able to track their learning progress for each course.
- A progress bar or percentage indicator must show overall course completion.
- Students must be able to view quiz/exam results, scores, and improvement recommendations.

Edit Profile

- Students must be able to update personal information such as **name, email, phone number, and profile photo**.
- Changes must be reflected immediately across the platform and saved securely.
- Users must be notified if email verification is needed after an email change.

Change Password

- Students must be able to securely change their password while logged in.
- The system must verify the current password before allowing a new one.
- After changing a password, the system must automatically log out all active sessions for account security.

Third: Teacher Side (Instructor Interface)

Dashboard Overview

- Teachers must be able to view a summary of their teaching activity immediately after login.
- The dashboard must display metrics such as **number of active courses**, **total enrolled students** and **recent quiz submissions**.
- Quick-access links must allow teachers to add new lessons, view assignments, or send announcements.

Course Management

- Teachers must be able to **create, edit, and delete** courses assigned to them.
- Each course must include key details: **course title, subject, description, and thumbnail image**.
- Teachers must be able to **organize lessons** within a course by topic or order.
- The system must allow teachers to **publish/unpublish** courses for students.

Lesson and Resource Upload

- Teachers must be able to upload and manage course materials such as **videos**, **PDF notes**, **presentations**, or **audio lectures**.

- Teachers must be able to edit or replace outdated files without affecting student progress.
- The platform must automatically optimize uploaded videos for multiple bandwidth levels (e.g., low/high quality).

Quiz and Exam Creation

- Teachers must be able to **create quizzes, mock exams, and assignments** within each course.
- Quizzes may include multiple question types: **multiple choice, true/false, and short answer**.
- Teachers must be able to define **time limits**, **passing scores**, and **feedback messages** for each assessment.
- Teachers must be able to view student performance statistics per quiz.

Student Progress Tracking

- Teachers must be able to view each student's progress in their courses.
- The system must display performance analytics such as **average scores** and **lesson completion rates**.
- Teachers must be able to export progress data for evaluation or reporting.

Messaging and Communication

- Teachers must be able to communicate directly with students through an in-app messaging system.
- Teachers must be able to send **announcements** to all students in a course (e.g., reminders, updates, or motivational messages).
- A notification system must alert teachers when they receive new messages.

Discussion Forum Moderation

- Teachers may pin important posts or FAQs to the top of discussion threads.

Live Sessions and Scheduling

- Teachers must be able to schedule live classes or webinars using an integrated video-conferencing link (e.g., Zoom, Google Meet).

- Students should receive automatic notifications when a session is scheduled.
- Attendance must be tracked and recorded automatically for reporting.

Profile Management

- Teachers must be able to update their profile details, including **name, specialization, biography, and profile photo**.
- Teachers may optionally link to professional profiles or credentials.
- Profile updates must be reviewed by the admin if moderation is enabled

Change Password

- Teachers must be able to securely change their password.
- Password change requests must require reauthentication.
- A confirmation email should be sent after every password update for security.

Analytics & Reports

- Teachers must be able to generate course-specific reports summarizing **enrollment trends, quiz statistics, and student engagement levels**.
- Reports should be exportable in **PDF or CSV** formats.
- The analytics dashboard should include simple visualizations such as charts or progress bars.

3.6 Non-Functional Requirements

Performance

- All pages (e.g., course listings, lesson pages, dashboards) must load within **3 seconds** under normal network conditions to ensure a smooth user experience.
- Video and PDF content should be optimized for low-bandwidth environments (e.g., adaptive streaming, compressed formats).
- The platform should efficiently handle concurrent access by multiple students during peak times, such as exam periods.

Security System

The platform boasts an advanced and comprehensive security system that includes:

• SQL Injection Protection: Use of PDO Prepared Statements and input sanitization.

• XSS Protection: Data sanitization and htmlspecialchars on all outputs.

• CSRF Protection: Tokenization system and request validation.

• Password Encryption: Use of Argon2ID (stronger than bcrypt).

• Session Protection: Secure sessions with HTTP and periodic renewal.

• Brute Force Protection: Account tracking and lockout after 5 failed attempts (for 15 minutes).

• Rate Limiting: Limiting 60 requests per hour to prevent automated attacks.

Usability

- The user interface must be **intuitive, consistent, and user-friendly**, accommodating users with different levels of digital literacy.
- The design must be **responsive**, adapting seamlessly to desktops, tablets, and smartphones.
- Clear navigation and visual feedback (e.g., progress bars, success/failure messages) must guide users through all learning activities.
- Accessibility standards (WCAG 2.1 Level AA) must be followed to support users with disabilities.

Availability

- The system must achieve **99.9% uptime annually**, excluding planned maintenance.
- Redundancy and failover mechanisms should be implemented to minimize downtime.
- Essential learning content (videos, PDFs) should remain accessible even during minor outages through caching and content delivery networks (CDNs).

Scalability

- The platform architecture must allow for **horizontal scaling** to support an increasing number of students, teachers, and uploaded courses.

- Database and media storage should be designed to handle large datasets (student records, lesson files, video content) without performance degradation.
- Cloud-based infrastructure (e.g., AWS or Google Cloud) should be leveraged
- for scalable deployment.

Maintainability

- The system codebase must be **modular and well-documented**, enabling easy maintenance and future enhancements.
- Version control (Git) must be used for collaborative development and rollback capability.
- Configuration files and environment variables must be externalized for easy deployment and updates.
- Automated testing and CI/CD pipelines must ensure reliable maintenance and upgrades without service interruption

3.6 Use Case Diagram

3.6.1 overview

The following use case diagram illustrates the main interactions between the three primary actors of the Himmah Virtual Learning Platform: **Student**, **Teacher**, and **Admin**.

Each actor interacts with system functionalities relevant to their role to support online learning, content management, and system administration.

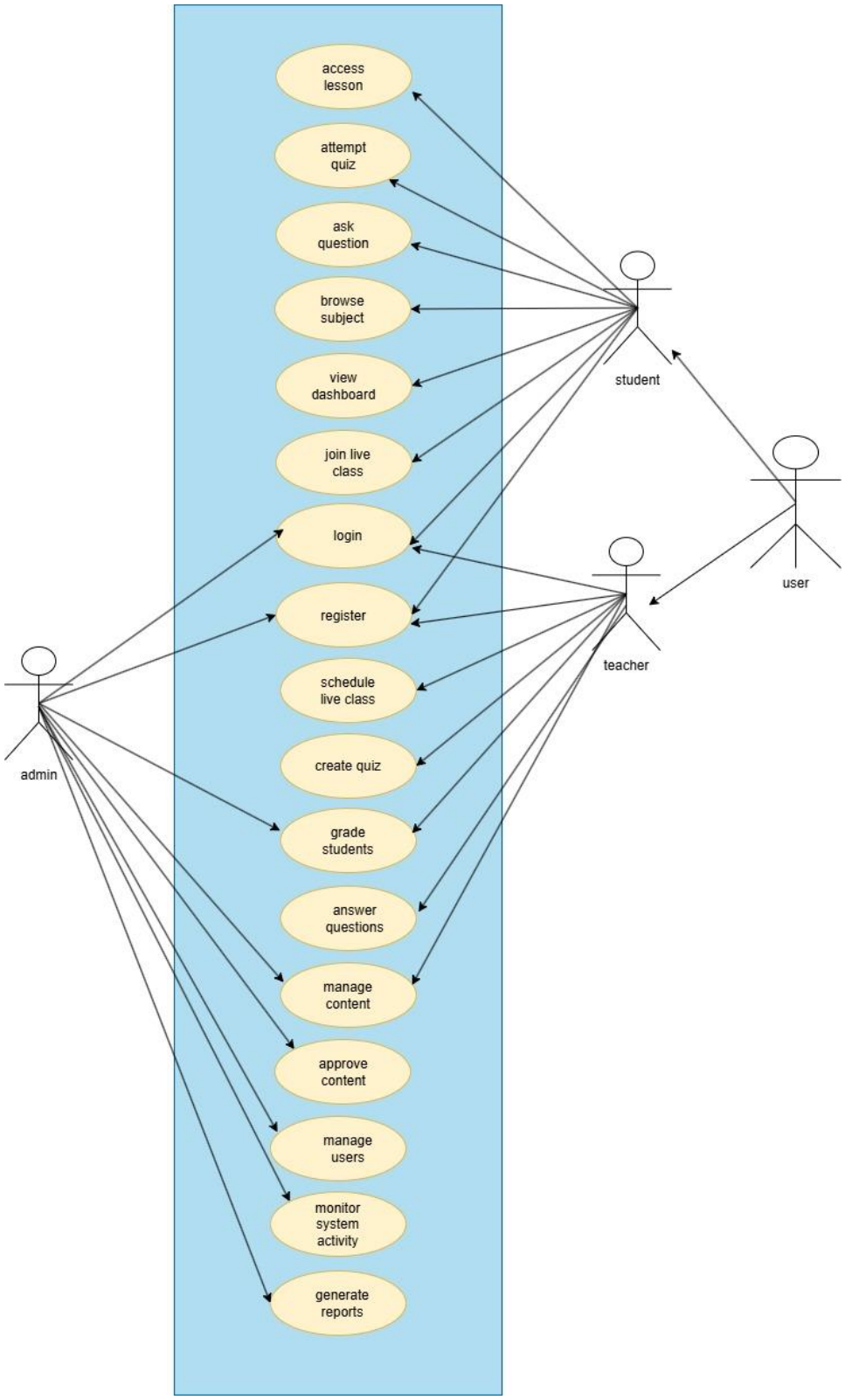

Figure 1 — Himmah Platform Use Case Diagram

3.6.2 Use Case Descriptions

A. Student Actor

- **Access Lesson:** View and study uploaded video lessons and PDF materials.
- **Attempt Quiz:** Take online quizzes and mock exams.
- **Ask Question:** Send inquiries or doubts to teachers via the messaging system.
- **Browse Subject:** Explore subject categories and available courses.
- **View Dashboard:** Track progress, quiz scores, and completed lessons.
- **Join Live Class:** Attend scheduled live sessions using integrated video tools.
- **Login / Register:** Create a new account or log in to access learning features.

B. Teacher Actor

- **Login / Register:** Create or access a teacher account.
- **Schedule Live Class:** Organize live video sessions for enrolled students.
- **Create Quiz:** Prepare quizzes and mock exams for their courses.
- **Grade Students:** Review quiz results and assign grades.
- **Answer Questions:** Respond to students' questions or messages.
- **Manage Content:** Upload and update lessons, videos, and course materials.

C. Admin Actor

- **Manage Content:** Review and organize platform-wide courses and lessons.
- **Approve Content:** Validate and publish teacher-submitted content.
- **Manage Users:** Add, update, or remove students, teachers, and admins.
- **Monitor System Activity:** Track overall system usage, activity logs, and performance metrics.
- **Generate Reports:** Produce analytical and performance reports for platform evaluation.

3.7 Use Case Scenarios

Admin Side Scenarios

Scenario 1: Manage Admin Accounts

User: Admin
Goal: Add, edit, or delete admin accounts
Steps:

1. Admin logs into the system control panel.
2. Navigates to the "Admin Accounts" section.
3. Views the list of all current admin users.
4. Adds a new admin by entering required details (name, email, role, and permissions).
5. Edits or deletes existing admin accounts as needed.

Outcome: Admin accounts are successfully managed with accurate permissions and updated information.

Scenario 2: Manage Homepage Slider

User: Admin
Goal: Upload, update, or arrange homepage slider images (e.g., announcements, new courses).
Steps:

1. Admin accesses the "Slider Management" section.
2. Uploads new promotional or educational banner images.
3. Sets display priority for each image (e.g., featured first).
4. Edits or deletes existing slider items as necessary.

Outcome: The homepage slider displays current and correctly ordered educational banners.

Teacher Side Scenarios

Scenario 1: Create and Manage Quizzes

User: Teacher
Goal: Create, edit, or delete quizzes for students.
Steps:

1. Teacher logs into the system.
2. Navigates to the "Quiz Management" section.
3. Creates a new quiz by entering questions, options, and correct answers.
4. Saves the quiz and assigns it to a specific subject or class.
5. Updates or deletes quizzes as required.

Outcome: Quizzes are created and made available to students for assessments.

Scenario 2: Schedule Live Class

User: Teacher
Goal: Organize and schedule a live online class session.
Steps:

1. Teacher logs into the platform.
2. Sends a message to students public group.
3. Sets the class date, time, subject, and meeting link.
4. Sends notifications to enrolled students.
5. Conducts the class and marks attendance.

Outcome: A live class is successfully scheduled and conducted, enabling real-time learning.

Student Side Scenarios

Scenario 1: Register and Login

User: Student
Goal: Create an account and access the platform.
Steps:

1. Student opens the platform's registration page.
2. Enters personal details such as name, email, and password.
3. Confirms registration through an email or verification code.
4. Logs into the platform using registered credentials.

Outcome: The student successfully creates and accesses their account.

Scenario 2: Access Lessons and Attempt Quizzes

User: Student
Goal: Learn course material and test understanding through quizzes.
Steps:

1. Student logs into the dashboard.
2. Selects a subject to access available lessons.
3. Reads through lesson content or watches recorded sessions.
4. Attempts quizzes related to the lesson.
5. Submits answers and views results or feedback.

Outcome: The student accesses and completes lessons and quizzes effectively.

Scenario 3: Join Live Class and Ask Questions

User: Student
Goal: Participate in real-time classes and interact with teachers.
Steps:

1. Student logs in and navigates to the "Live Classes"
2. Views the schedule of upcoming sessions.
3. Joins a live class at the scheduled time.
4. Asks questions using the chat or Q&A feature.
5. Receives responses or clarifications from the teacher.

6. **Outcome:** The student engages in interactive learning and receives real-time support.

Chapter 4

Bibliography

Bibliography

1. Hassan, S., & Shteiwi, M. (2020). *The Role of E-Learning in Enhancing Education in Gaza during the COVID-19 Pandemic*. *Educational Technology Research and Development*, 68(5), 2299-2315.
2. El-Sabbagh, M., & Al-Saadi, S. (2019). *Challenges and Opportunities of E-Learning in Gaza*. *International Journal of Educational Development*, 65, 102-110.
3. Johnson, R., & Brown, P. (2019). "Benefits of Interactive Learning in Online Platforms."
4. Harrison, T., & Taylor, S. (2020). "AI in Educational Content Personalization."
5. Liu, X., & Wang, Y. (2021). "Multimedia Learning for Secondary Education."
6. Lee, J., et al. (2021). "Improving Educational Website Design for High School Students."